Weekly Report – Week 6 – Yongchun Li (1378156)

1. Completed Work(≈ 6 hours)

https://github.com/SWEN90017-2025-SM1/OAS-web-based-oral-assessments/tree/Feat-3.2-AssessmentContex

https://github.com/SWEN90017-2025-SM1/OAS-web-based-oral-assessments/commit/d69f26dc3e19a0b0ca5e63c5d3eb95f57c954661

1. Story Mapping & Traceability: Mapped stories 3.1, 3.2, 6.6; clarified correct linkage of story 3.2 to the "Get session context" specification (section 4.2) and identified unimplemented assessment persistence (future 6.6 scope).
2. Session Context Endpoint: Implemented `GET /api/marking/sessions/{sessionId}/context` (branch: Feat-3.2-AssessmentContex) delivering a consolidated payload (assignment, session + time slot, participants, rubric, predefined comments, placeholders for assessment + venue enrichment).
3. Structured DTO Model: Designed `MarkingSessionContextDTO` with hierarchical rubric (criteria → levels) and auxiliary flattened predefined comment list; introduced deterministic transient IDs to support stable client state management.
4. Rubric Parsing & Normalisation: Single-pass JSON parsing (meta + array) producing denormalised representation while preserving order; resilient to absent optional comment arrays; future-ready for caching.
5. Error Handling Unification: Extended `BusinessError` with `EXAMINER_NOT_ASSIGNED` and replaced ad-hoc string messages, enabling consistent API error semantics and cleaner assertions.
6. Logging & Filter Fix: Corrected defect where business exceptions were logged as HTTP 200; now proper status propagation (commit d69f26d) ensuring observability & audit fidelity.
7. Technical Presentation: Prepared and delivered an interim internal presentation (slides + live walkthrough) demonstrating the new context endpoint, error code standardisation, and rubric modelling approach; gathered feedback on assessment persistence strategy.

2. Key Technical Challenges

1. Multi-Entity Aggregation: Resolved complexity via layered helper methods to isolate lookup concerns (session, assignment, participants, rubric) improving testability.
2. Heterogeneous Rubric JSON: Normalised mixed meta/array schema; generated stable IDs to avoid client re-diff churn.
3. Dual-Use Comments: Extracted predefined comments both in-place (levels) and as a flat catalogue without extra traversal (O(n)).
4. Error & Logging Semantics: Eliminated silent 200-on-error pathway in filter/aspect chain; unified code-based exceptions for analytics alignment.
5. Forward Extensibility: Preserved contract stability with placeholders rather than speculative fields (avoiding future breaking changes).

3. Plan for Next Week

1. Implement `existingAssessment` retrieval (draft/submitted) once schema finalised; design idempotent upsert contract for story 6.6 foundation.
2. Authorisation Hardening: Introduce examiner ownership checks (method-level security) post global security rule refinement.
3. Rubric Applies-To Logic: Derive individual/group applicability instead of current default (add schema flag or inference heuristic).
4. Caching Strategy: Prototype per-assignment rubric cache (in-memory or Caffeine) and benchmark deserialisation savings.
5. Integration Tests: Add end-to-end test with in-memory DB covering group participant expansion and predefined comments integrity.
6. Venue Enrichment: Populate building/room or virtual channel fields once upstream schema confirmed.
7. Developer Guide Addendum: Publish example session context JSON with glossary for front-end consumption.